🚀R&D DOCUMENT ON THREE-TIER AZURE ARCHITECTURE DEPLOYMENT WITH VMs

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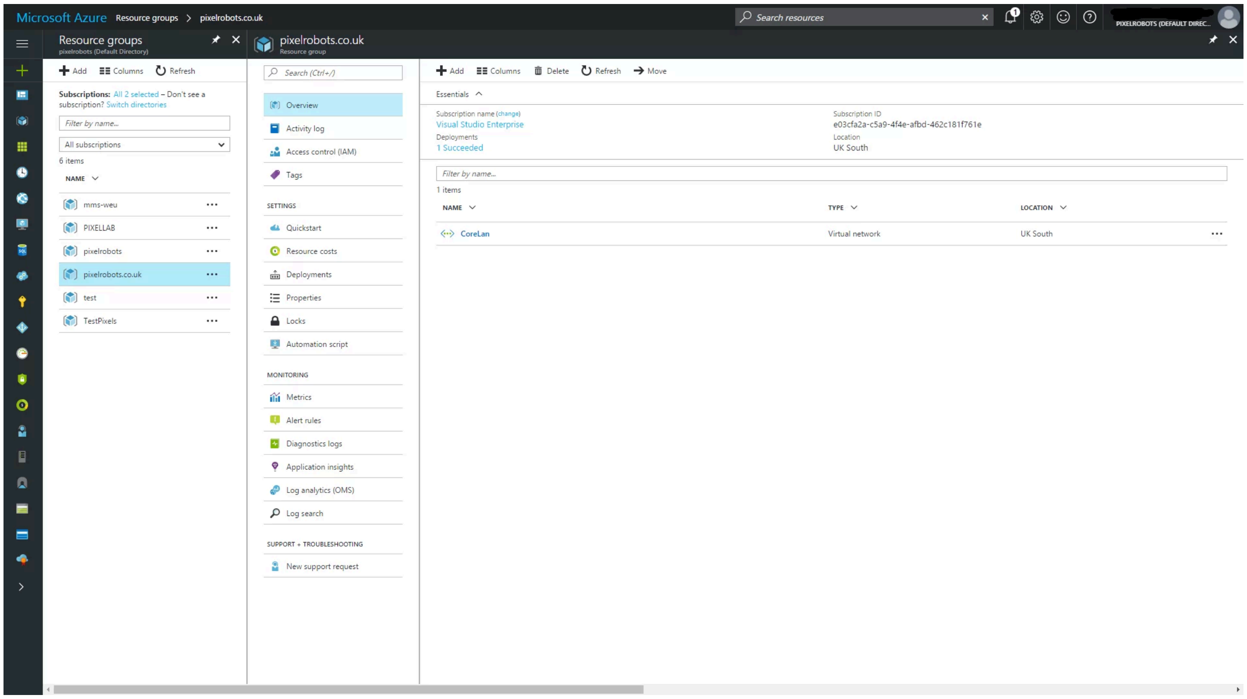
PURPOSE :
This document outlines the process of deploying a three-tier architecture (Web Tier, App Tier, and DB Tier) in Azure. It demonstrates network segmentation using subnets and controls access through Network Security Groups (NSGs).

🔎ARCHITECTURAL DEPLOYMENT

Step 1: Sign In to Azure Portal

- Navigate to https://portal.azure.com.
- Log in with your Azure credentials.

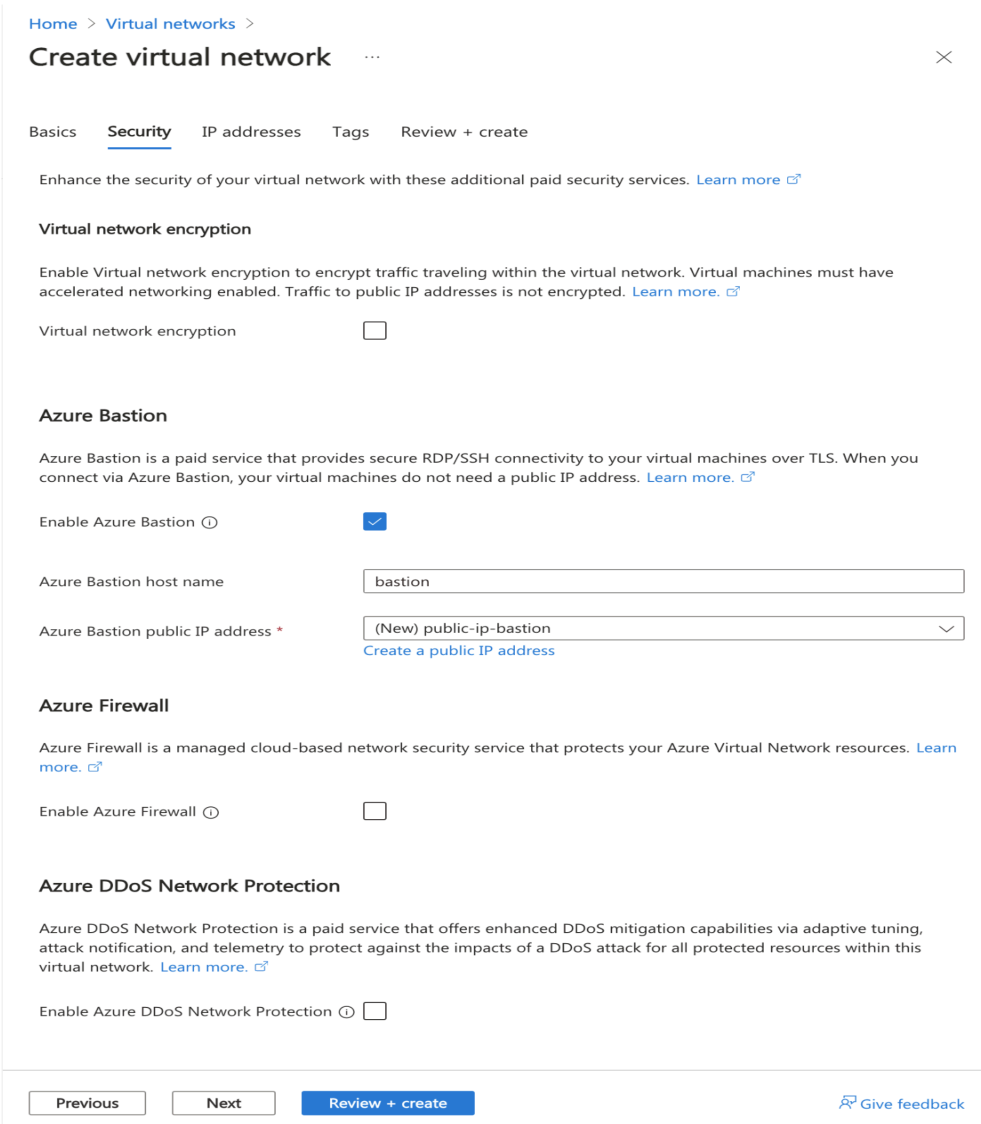
Step 2: Create a Resource Group

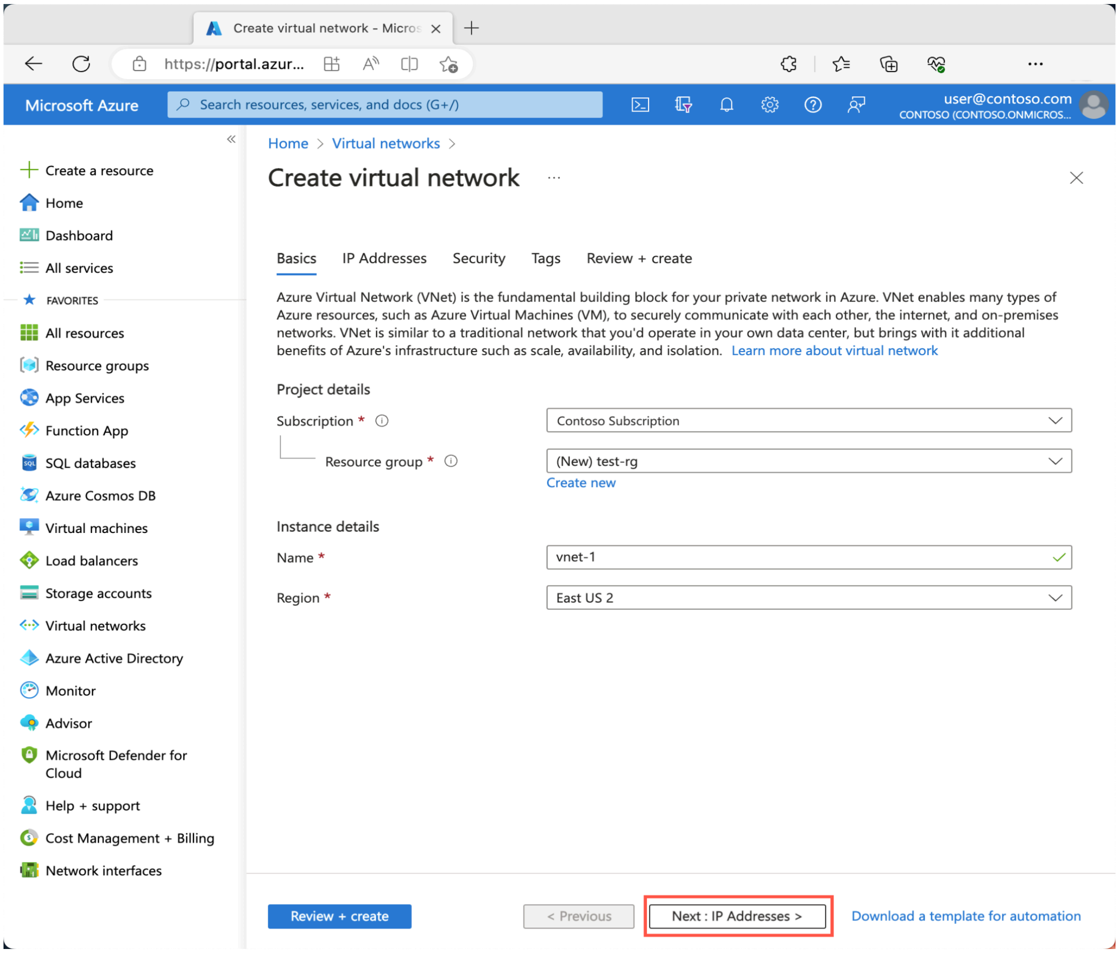


Steps:

1. Search for “Resource groups” in the search bar → Click Create.
2. Enter the Resource Group name: ThreeTier-RG.
3. Choose your Region (e.g., East US).
4. Click Review + Create → Create.

Step 3: Create a Virtual Network (VNet)

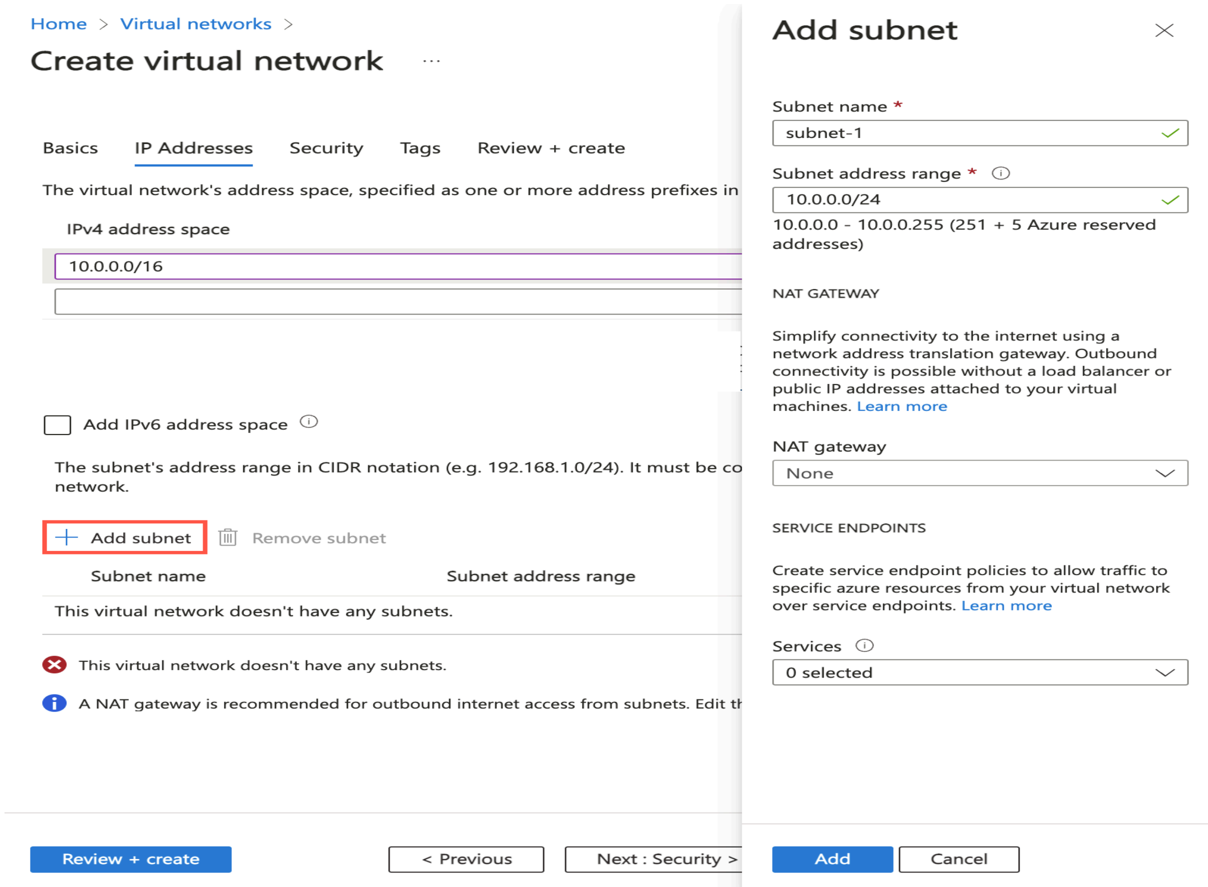

Home > Virtual networks >
Create virtual network
Basics
Security
IP addresses
Tags
Review + create
Enhance the security of your virtual network with these additional paid security services. Learn more
Virtual network encryption
Enable Virtual network encryption to encrypt traffic traveling within the virtual network. Virtual machines must have accelerated networking enabled. Traffic to public IP addresses is not encrypted. Learn more.
Virtual network encryption
Azure Bastion
Azure Bastion is a paid service that provides secure RDP/SSH connectivity to your virtual machines over TLS. When you connect via Azure Bastion, your virtual machines do not need a public IP address. Learn more.
Enable Azure Bastion
Azure Bastion host name
bastion
Azure Bastion public IP address *
(New) public-ip-bastion
Create a public IP address
Azure Firewall
Azure Firewall is a managed cloud-based network security service that protects your Azure Virtual Network resources. Learn more.
Enable Azure Firewall
Azure DDoS Network Protection
Azure DDoS Network Protection is a paid service that offers enhanced DDoS mitigation capabilities via adaptive tuning, attack notification, and telemetry to protect against the impacts of a DDoS attack for all protected resources within this virtual network. Learn more.
Enable Azure DDoS Network Protection
Previous
Next
Review + create
Give feedback



Steps:

1. Search for “Virtual networks” → Click Create.
2. Resource Group: Select ThreeTier-RG.
3. Name: ThreeTier-VNet.
4. Region: Same as the Resource Group.
5. IPv4 Address space: 10.0.0.0/16.
6. Click on Next: IP Addresses.

Step 4: Create Subnets



4.1. Create WebSubnet

- Click Add Subnet
    - Subnet Name: WebSubnet
    - Subnet Address Range: 10.0.1.0/24
    - Click Add

4.2. Create AppSubnet

- Click Add Subnet
    - Subnet Name: AppSubnet
    - Subnet Address Range: 10.0.2.0/24

- Click Add

4.3. Create DbSubnet

- Click Add Subnet
  - Subnet Name: DbSubnet
  - Subnet Address Range: 10.0.3.0/24
  - Click Add

Step 5: Review and Create VNet

- Click Review + Create.
- Click Create.

Your VNet with three subnets is now ready.

Step 6: Verify Subnets

1. Go to your created VNet (ThreeTier-VNet).
2. Click on Subnets.
3. You should see all three subnets listed:

🔎**DEPLOYING VM IN EACH SUBNET**

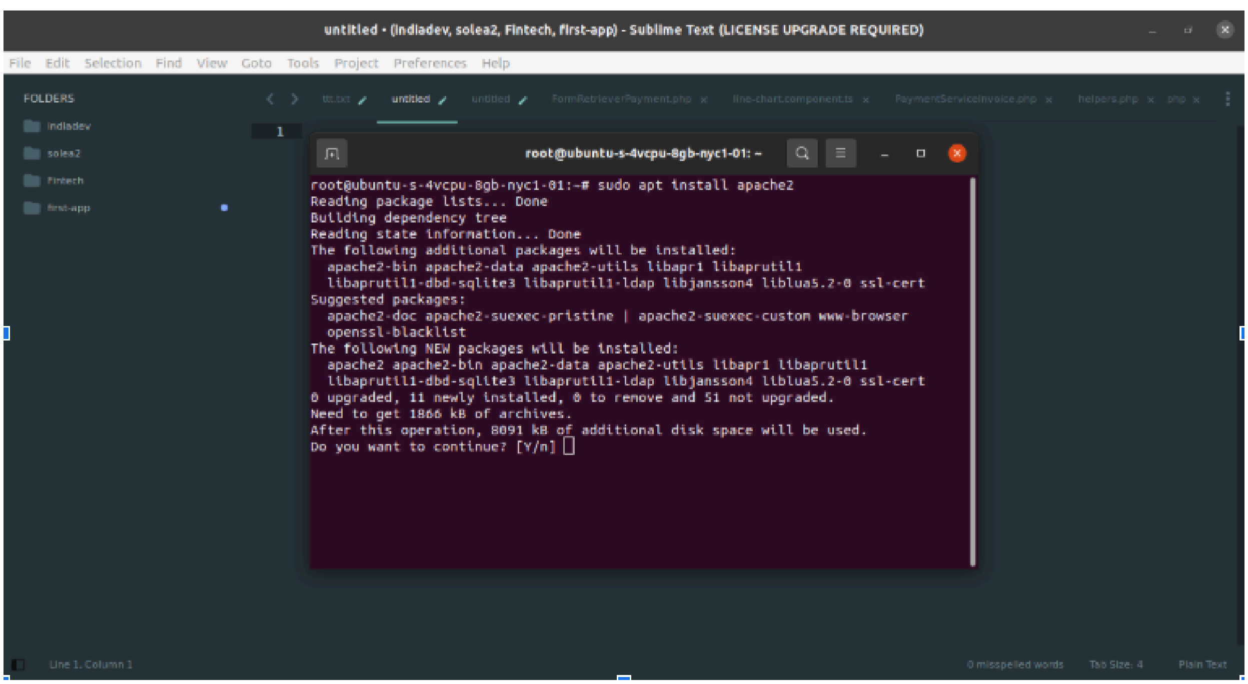
VM Creation Steps (Repeat for each VM)

1. Search “Virtual Machines” → Create
2. Choose:

    - Resource Group: ThreeTier-RG
    - VM Name: <Your VM Name>
    - Image: Ubuntu 20.04 LTS / Windows Server 2019
    - Size: Standard_B2s (or per your need)
    - Authentication: SSH for Linux, Password/RDP for Windows
    - Public IP: Only for WebSubnet VMs
    - NIC → Select correct subnet (WebSubnet, AppSubnet, DbSubnet)
    - Assign NSG: No (subnet-level NSG is already configured)
3. Click Review + Create

Apache Setup on Linux VMs

SSH into Linux VMs:

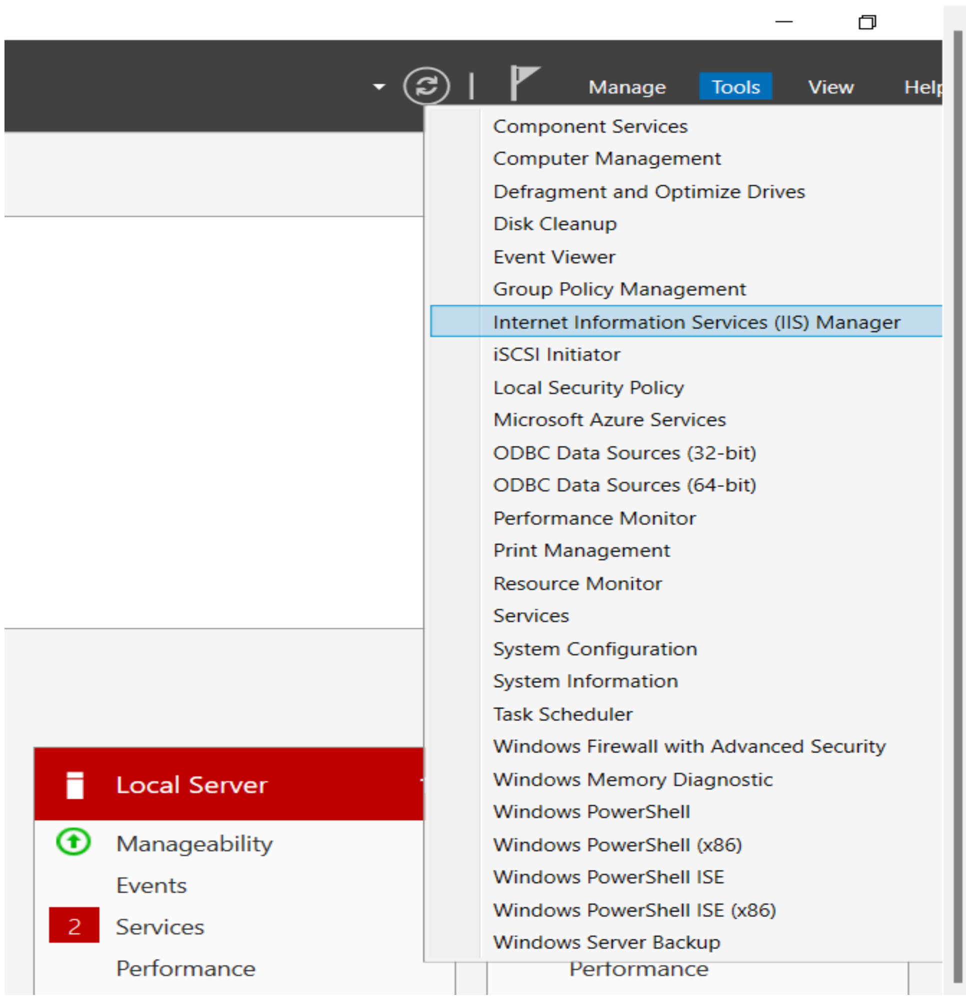


Test: Open a browser → http://<Public IP of Web-Linux-VM>
(Default Apache page should load)

IIS Setup on Windows VMs

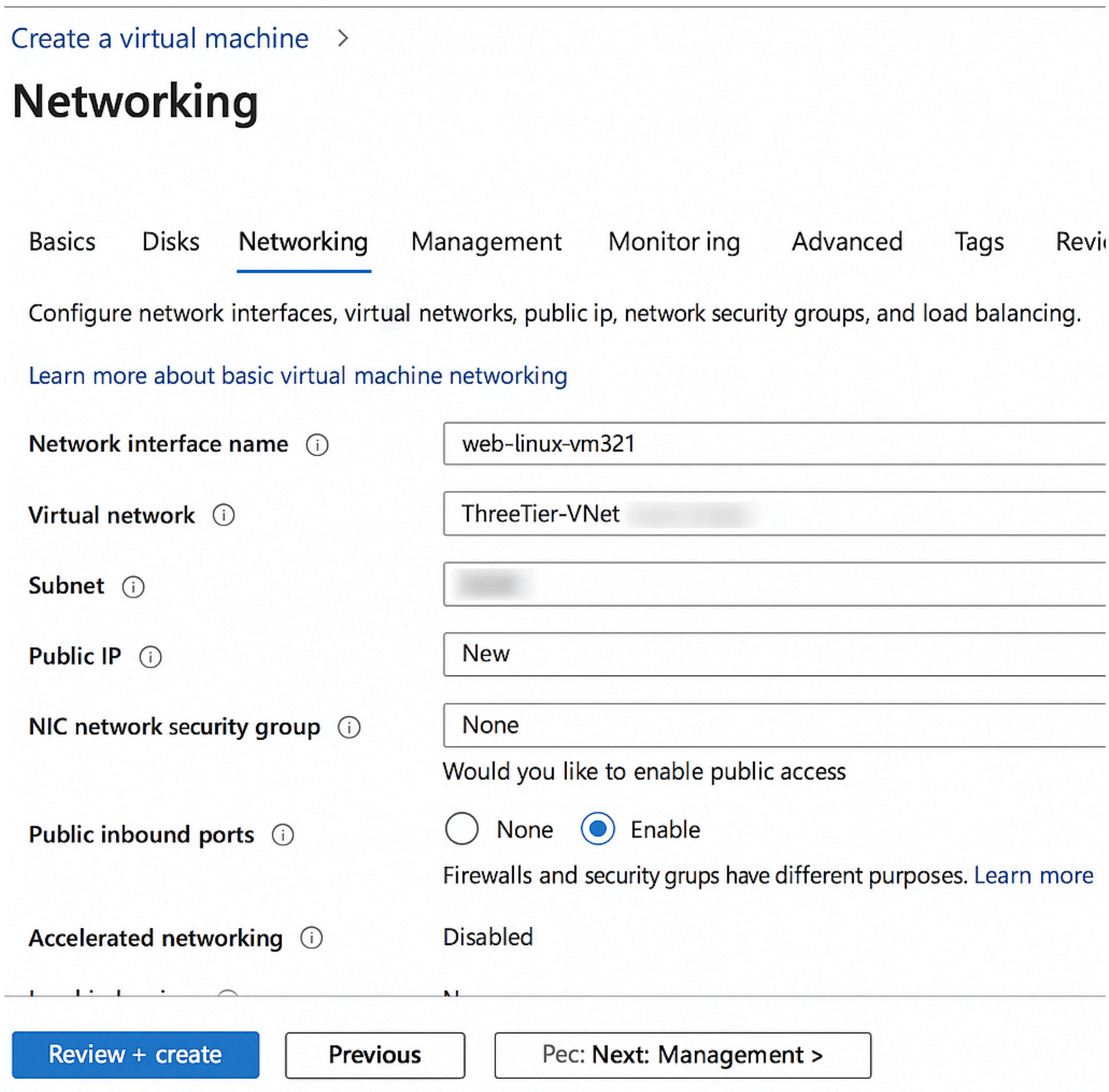
RDP into Windows VMs:

- Open PowerShell as Administrator



Test: Open browser → http://localhost or http://<Public IP of Web-Windows-VM>

Test Tier Communication



Next steps after configuration :

1. Complete the wizard for each VM:
    - Web tier: 1 Linux + 1 Windows
    - App tier: 1 Linux + 1 Windows
    - DB tier: 1 Linux + 1 Windows

2. Under Networking, ensure correct subnet selection, public IP assignment (if needed), and inbound ports.
3. Click Review + create → Create.
4. Repeat for all six VMs, adjusting names and subnet as above.
